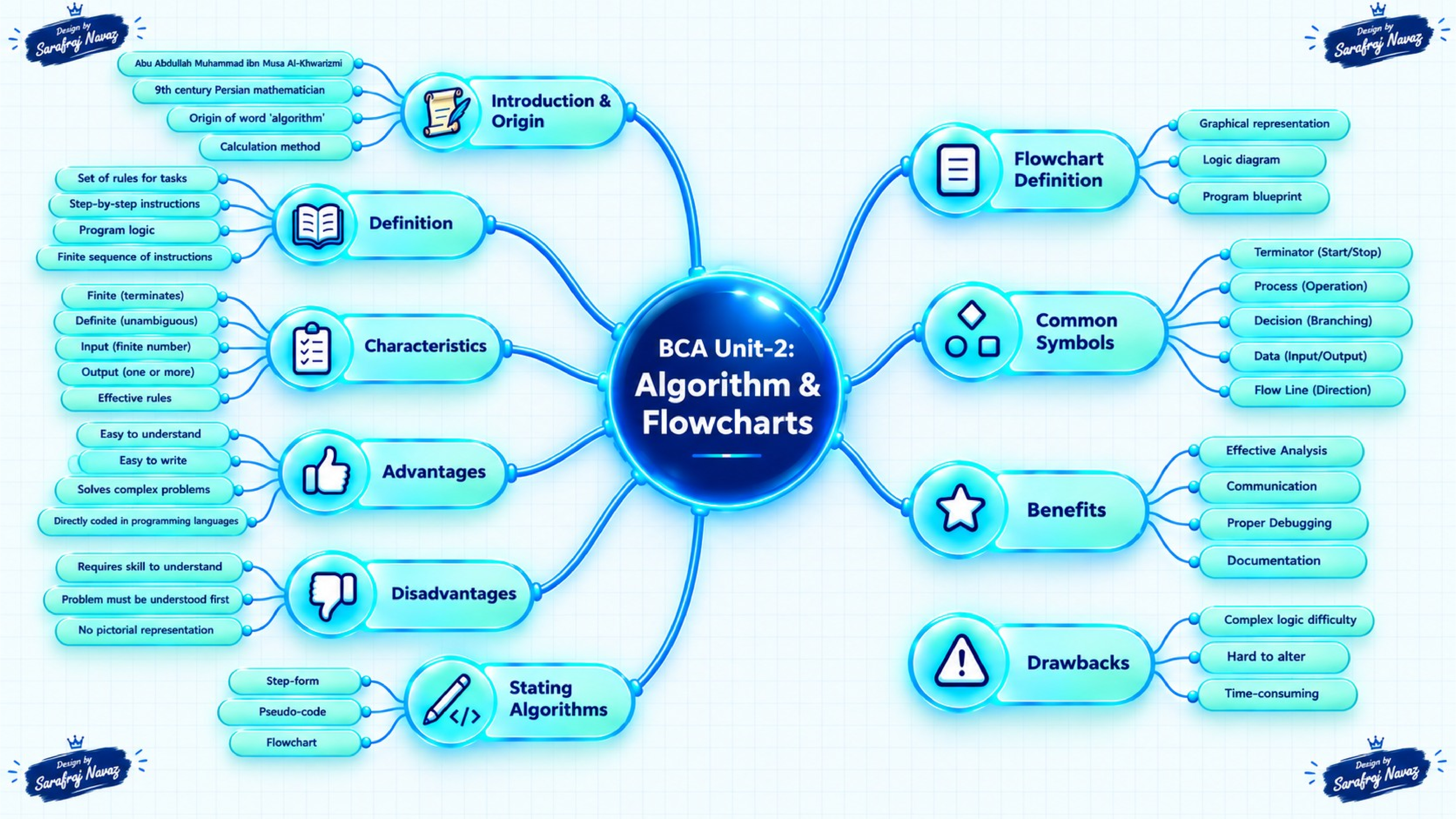

BCA Unit-2: Algorithm & Flowcharts
Introduction & Origin
Abu Abdullah Muhammad ibn Musa Al-Khwarizmi
9th century Persian mathematician
Origin of word 'algorithm'
Calculation method
Definition
Set of rules for tasks
Step-by-step instructions
Program logic
Finite sequence of instructions
Characteristics
Finite (terminates)
Definite (unambiguous)
Input (finite number)
Output (one or more)
Effective rules
Advantages
Easy to understand
Easy to write
Solves complex problems
Directly coded in programming languages
Disadvantages
Requires skill to understand
Problem must be understood first
No pictorial representation
Stating Algorithms
Step-form
Pseudo-code
Flowchart
Flowchart Definition
Graphical representation
Logic diagram
Program blueprint
Common Symbols
Terminator (Start/Stop)
Process (Operation)
Decision (Branching)
Data (Input/Output)
Flow Line (Direction)
Benefits
Effective Analysis
Communication
Proper Debugging
Documentation
Drawbacks
Complex logic difficulty
Hard to alter
Time-consuming
Design by Sarafraj Navaz

2.1 ALGORITHM

Introduction & Origin

Introduction

The word "algorithm" is the name of Abu Abdullah Muhammad ibn Musa Al-Khwarizmi who was a Persian mathematician in 9th century. An algorithm is a set of rules which can be used to do some task either manually or electronically (computer system) is known as an algorithm. Most commonly, this term is used with reference to computers. Generally algorithms are used to provide solutions of different problems. Algorithms provide effective methods for problem solving. It solves a problem with a finite sequence of instructions.

Origin

The term "algorithm" is derived from the name of the great Persian mathematician Abu Abdullah Muhammad ibn Musa Al-Khwarizmi, who wrote a book on algebra in the 9th century. His name is often considered the origin of the word "algorithm".

Key Points

- Algorithm is the name of Abu Abdullah Muhammad ibn Musa Al-Khwarizmi.
- It is a set of rules to solve a task.
- Commonly used in computers for problem solving.
- Provides effective methods and finite sequence of instructions.

2.1.2 Meaning and Definition

Definition

The term "Algorithm is taken from a Latin word "algorismus". The meaning of this word is "calculation method". To solve a problem specific set of instructions are used, this set of instruction is known as algorithm. For a given problem it is a step by step instruction for how to reach at the solution of that problem. Basically algorithm shows the logic of the program. An algorithm is defined as a finite set of well defined instructions for problem solving. It takes a valid input and produces desired output in a finite amount of time. It is illustrated in **figure 2.1.**

Key Points

- An algorithm is generally used for calculation and processing data using a sequence of finite instructions.
- An algorithm provides step-by-step procedure for problem solving.
- According to Stone and Knuth "an algorithm is a set of rules that precisely defines a sequence of operations in such a way that each rule is effective and definite and that the sequence terminates in a finite time." An algorithm has a specific format which is shown in figure 2.2.

Figure 2.1: Algorithm

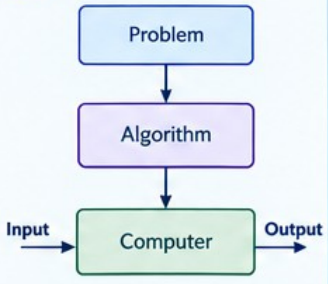


Figure 2.1: Algorithm

An algorithm is generally used for calculation and processing data using a sequence of finite instruction set.

Disadvantages of Algorithms

- Occupy more skill to understand it.
- Problem should be understand before proceed for algorithm.
- Cannot be represented in pictorial form.

Examples

Example 1: Find the area and circumference of a circle of radius r.

Algorithm

Step 1: Read Radius of Circle: as R

Step 2: Assign Value of PI = 3.14

Step 3: Calculate Area = PI * R * R

Step 4: Calculate Circumference Circum = 2 * PI * R

Step 5: Print Value of Area and Circum

Step 6: END / STOP

C Program

```c
#include <stdio.h>
#include <math.h>
int main() {
    float R, PI = 3.14, Area, Circum;
    printf("Enter radius of circle: ");
    scanf("%f", &R);
    Area = PI * PI * R;
    Circum = 2 * PI * R;
    printf("Area = %.2f\n", Area);
    printf("Circumference = %.2f\n", Circum);
    return 0;
}
```

Output

```
Enter radius of circle: 5
Area = 78.50
Circumference = 31.40
```

Explanation

The program takes the radius as input, calculates the area and circumference using the given formulas and displays the result.

Example 2

Find the area and circumference of a circle of radius r.

Algorithm

Step 1: Read Radius of Circle as R

Step 2: Assign Value of PI = 3.14

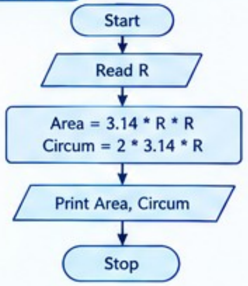
Step 3: Calculate Area = PI * R * R

Step 4: Calculate Circumference Circum = 2 * PI * R

Step 5: Print Value of Area and Circum

Step 6: END / STOP

Flowchart



C Program

```c
#include <stdio.h>
#include <math.h>
int main() {
    float R, PI = 3.14, Area, Circum;
    printf("Enter radius of circle: ");
    scanf("%f", &R);
    Area = PI * R * R;
    Circum = 2 * PI * R;
    printf("Area = %.2f\n", Area);
    printf("Circumference = %.2f\n", Circum);
    return 0;
}
```

Output

```
Enter radius of circle: 5
Area = 78.50
Circumference = 31.40
```

Explanation

The program takes the radius as input, calculates the area and circumference using the given formulas and displays the result.

ALGORITHMS & FLOWCHARTS

Examples 4 – 7

C Example 4: Calling a friend on the telephone.

Algorithm

Step 1: Pick-up the phone (receiver).
Step 2: Dial the friend's number (Phone Number).
Step 3: If busy, hang-up phone, wait if friend does not reply then jumps to step 2.
Step 4: If no one answers, leave a message, keep down the receiver.
Step 5: STOP

Explanation

The program takes the friend's phone number and the user's choice. If the friend is busy or does not reply, the program asks the user to redial. If no one answers, the user can leeve a message. Finally, the call ends and the receiver is kept down.

</> C Program

```c
#include <stdio.h>
#include <conio.h>
int main() {
    char choice;
    int number;
    printf("Enter friend's phone number: ");
    scanf("%d", &number);
    printf("If busy or no answer,\n");
    printf("1. Hang up and redial\n");
    printf("2. Leave message\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);
    if (choice == 1){
      printf("Hang up and redial the number.\n");
    } else if(choice == 2) {
      else if (Choice a message.\n");
    } else {
      printf("Invalid choice.\n");
    }
    printf("Call ended. Keep down the receiver.\n");
    return 0;
}
```

C Example 5: Write the steps for calculating the average of two numbers.

Algorithm

Step 1: START
Step 2: Read the first number.
Step 3: Read the second number.
Step 4: Add both numbers.
Step 5: Divide the sum by 2.
Step 6: Display the average.
Step 7: STOP

Explanation

The program reads two numbers, adds them and divides the sum by 2 to find the average. The result is then displayed on the screen.

</> C Program

```c
#include <stdio.h>
int main() {
    int num1, num2;
    float avg;
    printf("Enter first number: ");
    scanf("%d", &num1);
    printf("Enter second number: ");
    scanf("%d", &num2);
    avg = (num1 + num2) / 2.0;
    printf("Average = %.2f\n", avg);
    return 0;
}
```

C Example 6: Write the steps for calculating the simple interest.

Algorithm

Step 1: Read principal, rate and time.
Step 2: Calculate I = P × R × T.
Step 3: Divide the result by 100.
Step 4: Display S.
Step 5: STOP

Explanation

The program takes principal, rate and time as input, calculates the simple interest using the given formula and displays the result.

</> C Program

```c
#include <stdio.h>
int main() {
    float P, R, T, S;
    printf("Enter principal: ");
    scanf("%d", &P);
    printf("Enter rate: ");
    scanf("%d", &R);
    printf("Enter time: ");
    scanf("%d", &T);
    S = (P * R * T) / 100;
    printf("Simple Interest = %.2f\n", S);
    return 0;
}
```


C Example 7: Write the steps for finding the largest of three numbers.

Algorithm

Step 1: Read first number.
Step 2: Read second number.
Step 3: Read third number.
Step 4: Compare and find the largest.
Step 5: Display the largest number.
Step 6: STOP

Explanation

The program reads three numbers and compares them using conditional statements to find the largest number. The result is then displayed on the screen.

</> C Program

```c
#include <stdio.h>
int main() {
    int a, b, c, largest;
    printf("Enter first number: ");
    scanf("%d", &a);
    printf("Enter second number: ");
    scanf("%d", &b);
    printf("Enter third number: ");
    scanf("%d", &c);
    largest = a;
    if (b > largest) largest = b;
    if (c > largest) largest = c;
    printf("Largest number is: %d\n", largest);
    return 0;
}
```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